Exposure Planner: From Transient Candidate to Physical Exposure Setup

CRITO / TDMMA–2026 Workshop

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Purpose

This guide explains how to use the CRITO exposure planner for transient follow-up and how to turn its numbers into an observing decision. The planner is a calculator and a safety check; the final exposure is chosen by combining signal-to-noise, saturation, cadence, readout overhead, and the scientific goal.

1 The final workflow

1. **Select a transient.** In *Transients*, inspect the candidate, coordinates, magnitude, class, and visibility. Choose *Plan observation*. Do not send the transient directly to the queue.
2. **Open the plan.** CRITO creates a draft in *Plans* and pre-fills the candidate name, coordinates, and target magnitude.
3. **Enter the observing setup.** Check focal length, pixel size, seeing, filter, gain, detector temperature, and (if known) brightest-star magnitude. Set the required total SNR and the optional maximum sub-exposure.
4. **Compute the exposure.** Use the embedded Exposure Planner. It reports the sky/read-noise floor, saturation ceiling, SNR of one sub, number of subs, total integration, and the expected final SNR.
5. **Choose the science exposure.** Start from the planner's valid window, then choose a practical sub-exposure that gives useful cadence and preferably a measurable individual transient (often about 5–10 sigma when the light curve requires per-frame detections). Never exceed the saturation ceiling.
6. **Apply to recipe.** Press *Compute & apply to recipe*. The filter, gain, exposure time, and number of subs populate the first light recipe. Then set binning, dither, autofocus, centering, timing, and any repeats.
7. **Review and commit.** Save the plan. Use *Send to queue* for normal execution or *Run now* for immediate execution. Scheduling is done from the plan, not from the transient list.

2 What the planner is calculating

For a source of magnitude m, the calibrated source rate is approximately

$$S = ZP\,10^{-0.4m} \quad [e^-\,s^{-1}],$$

where ZP is the photometric zero point. In a sub-exposure of length t, source electrons are St.

The angular scale of one detector pixel is

$$p = 206.265\,\frac{\text{pixel size }(\mu\text{m})}{\text{focal length (mm)}} \quad [\text{arcsec}\,\text{pixel}^{-1}].$$

Seeing in pixels is $F = \text{seeing}/p$. The planner uses a seeing-dependent circular aperture, with area n_{pix} pixels. The larger the seeing or the larger the chosen aperture, the more sky noise is included.

For a simple aperture-photometry estimate, the variance in one frame is

$$\sigma^2 = St + n_{\text{pix}}(Bt + Dt + R^2) + n_{\text{pix}}\,\sigma^2_{\text{sky}},$$

where B is sky rate per pixel, D is dark-current rate per pixel, R is read noise per pixel per read, and the final term represents uncertainty in estimating the sky background. The single-frame signal-to-noise ratio is

$$\text{SNR}_1 = \frac{St}{\sqrt{\sigma^2}}.$$

For N independent equal exposures, approximately

$$\text{SNR}_{\text{total}} \simeq \sqrt{N}\,\text{SNR}_1, \qquad N = \left\lceil \left(\frac{\text{SNR}_{\text{goal}}}{\text{SNR}_1}\right)^2 \right\rceil.$$

The total integration is Nt. Camera readout, dithers, autofocus, and other overheads are added to the wall-clock time in the real observation.

3 How the inputs are correlated

Input	Direct effect	Practical consequence
Target magnitude m	Fainter by 1 mag means S is divided by 2.512.	A faint target needs longer subs or many more subs; check whether the requested SNR is realistic.
Required SNR	Sets the total signal precision. $N \propto \mathrm{SNR}^2_{\mathrm{goal}}$.	Doubling the goal requires about four times the total integration.
Focal length	Changes plate scale p.	Longer focal length gives smaller pixels on the sky, more pixels across the PSF, and usually a larger aperture in pixels.
Pixel size	Changes p in the opposite direction to focal length.	Larger angular pixels may reduce the number of pixels in the aperture, but can undersample the PSF.
Seeing	Increases the PSF width and aperture area.	More sky pixels are summed, so sky noise rises; poor seeing generally requires more time.
Filter	Changes zero point, throughput, and sky background.	A faster filter raises source rate but may also raise sky; use the measured calibration for that filter.
Gain	Changes electrons per ADU and often the effective read-noise behavior.	It does not create photons. Use the calibrated gain and check saturation in ADU and electrons.
Sensor temperature	Changes dark current.	Warmer sensors add dark noise and may shorten the useful exposure window.
Brightest-star magnitude	Adds a field-star saturation constraint.	The brightest star, not only the transient, can set the maximum safe exposure.
Max sub (optional)	User-imposed upper limit.	Use it for cadence, tracking, cosmic-ray, or moving-target constraints; the planner then increases N if needed.

4 Understanding the outputs

Sky-limited floor The approximate exposure needed for read noise to be a small part of the sky noise. A common heuristic is $t_{\mathrm{floor}} = 10R^2/B$. It is a lower bound for efficiency, not the science exposure by itself.

Saturation ceiling The longest safe exposure before the target or a specified bright star reaches the detector's linearity/full-well limit. Always remain below this value, with margin.

Recommended sub The current planner's practical starting point inside the valid window. Treat it as a candidate, then adjust it for cadence and per-frame detectability.

SNR per sub The expected SNR in one exposure. If it is below 1, the target is not individually detected; stacking can still reach the requested total SNR, but the light curve cadence is weak.

Subs $\times$ sub length The number of frames and their exposure time. More short frames reduce the damage from a bad frame and permit finer cadence, but increase readout overhead and data volume.

Total integration $N \times t$, excluding overhead unless explicitly shown. Add readout, dither, autofocus, and slew time when planning the night.

Achieved SNR and magnitude error The predicted final precision. For small errors, $\sigma_m \approx 1.0857/\mathrm{SNR}$ magnitudes.

Source rate, sky rate, plate scale, FWHM, aperture Intermediate values that explain the result and help diagnose bad assumptions or uncalibrated constants.

5 How to choose short or long subs

Choose a **shorter** sub when saturation, rapid fading, motion, guiding risk, or a fast cadence is more important than per-frame SNR. Choose a **longer** sub when read noise is important and the target remains safely below saturation. A useful decision sequence is:

1. Reject any exposure below the sky/read-noise floor unless there is a specific cadence reason.
2. Reject any exposure above the saturation ceiling; keep a margin for seeing variations and imperfect centering.
3. Within the remaining window, ask whether one frame should detect the transient. If yes, increase t until the desired single-frame SNR (often 5–10) is reached, subject to saturation.
4. Compare cadence: $t_{\mathrm{cadence}} \approx t + t_{\mathrm{readout}} + t_{\mathrm{dither}}$.
5. Compute wall-clock time: $T_{\mathrm{wall}} \approx N(t + t_{\mathrm{overhead}})$. If it is too long, lower the SNR goal, use a faster filter, improve the aperture/seeing setup, or split the program across visits.

6 Worked example

Suppose the planner reports $m = 18$, target SNR 30, $S = 15.85\ e^-\ s^{-1}$, sky $B = 8\ e^-\ s^{-1}\ pixel^{-1}$, FWHM = 5.02 pixels, and $n_{\mathrm{pix}} = 177.8$. With read noise $R = 1.1\ e^-$, the sky floor is

$$t_{\mathrm{floor}} = \frac{10(1.1)^2}{8} = 1.5 \text{ s}.$$

A 3 s exposure is above that floor, but its predicted SNR is only about 0.7. It can reach SNR 30 after many frames, yet it is a poor choice if each light-curve point must be visibly detected. A longer exposure, for example 60–180 s if it remains below saturation, produces fewer frames and a more useful cadence. The exact choice must be recomputed with the calibrated detector and checked against the saturation ceiling.

Workshop sentence

"The planner converts the target magnitude and calibrated instrument setup into source and sky electrons, finds the safe exposure window between read-noise inefficiency and saturation, and then chooses enough sub-exposures to reach the requested total SNR. I use that result as a starting point, then select the longest safe exposure that gives the cadence and per-frame detection needed by the science."